

QArchGauge: HPC-Driven Architecture Targeting for Application-Level Quantum Advantage

Abstract—Application-level quantum advantage requires a complete circuit application to beat a strong native HPC path on the same input and at the same output quality. Simulator throughput and isolated fault-tolerant resource counts do not reveal which QPU resource can move that boundary. We present QArchGauge, an HPC-driven architecture diagnosis framework that first gates every paired record by direct finite-shot application quality and then inverts its native deadline through workload-valid schedules, case-level reliability, physical space, and coupled resource improvements. Perlmutter supplies 3,552 ML, chemistry, optimization, and simulation records spanning 222 structural configurations and 16 seeds per configuration; its GPU time is evidence cost, never a QPU clock. At the declared noiseless tolerance, ML and optimization have no passing records, while 48/224 chemistry and 256/512 simulation records pass. Direct 10,000-shot closure further leaves only 12 simulation records with a same-record, full-loop application contract; all other physical timings are conditional lower bounds. For these 12 records, a QDK-aligned 1% reliability contract selects distance 13–15 and 79–86 T states per arbitrary rotation. Magic-state supply must reach $39,351\text{--}42,961\times$ a 64-factory baseline, or 2.52–2.75 million factories, before another component can dominate; six records remain blocked by the non-factory floor. Across 384 deterministic joint-design scenarios, factory supply is the first target for all eligible records throughout the $1\text{--}100\times$ region, then shifts to shot parallelism or logical-cycle latency as supply increases. A matched-event LSQCA replacement provides no core-area benefit at 4–7 logical qubits and its load/store movement removes all six otherwise runtime-favorable cases. Leadership HPC thus acts as an empirical oracle for falsifiable first, crossover, and next QPU architecture targets rather than evidence of present quantum speed.

I. INTRODUCTION

Quantum computing targets machine learning, chemistry, optimization, and scientific simulation, but executing a circuit is not application-level quantum advantage [1]–[3]. The circuit application must beat a valid native HPC implementation on the same input and at the same output quality after encoding, repeated evaluations, measurement, classical optimization, and postprocessing. For an architect, the central problem is therefore to identify whether the limiting resource is algorithmic quality, logical execution, synthesized rotations, useful shot parallelism, decoder/control reaction, physical space, or the moving native deadline.

Leadership systems provide a way to study this boundary before large fault-tolerant machines exist. cuQuantum, ScaleQsim, and AURORA-Q expand the circuit scale and throughput that HPC can inspect [4]–[6]. Their performance is

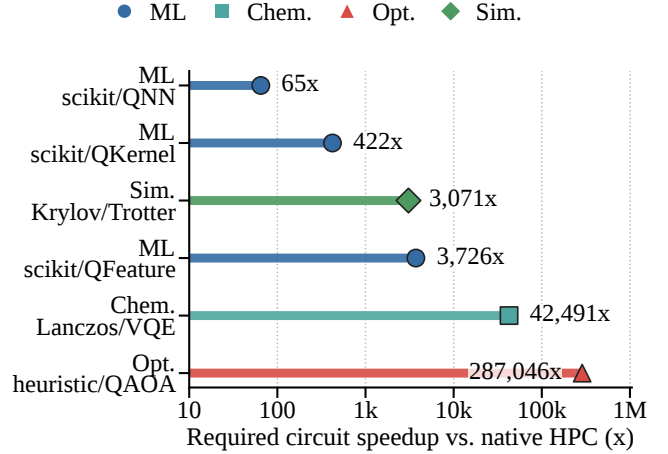


Fig. 1: Measured circuit-application speedup required to match each same-input native path. ML is classification, Chem. is molecular-energy estimation, Opt. is combinatorial optimization, and Sim. is Hamiltonian simulation. These ratios measure simulator-backed evidence cost, not projected QPU time.

not future-QPU performance. A faster state-vector kernel can leave the complete circuit path behind a native solver, and it says nothing about code distance, finite shots, factory space, or output quality. A defensible study must therefore separate measured evidence cost, $T_{\text{qsim}}/T_{\text{native}}$, from projected hardware cost, $T_{\text{qhw}}/T_{\text{native}}$, while keeping both attached to the same application record.

Figure 1 exposes the problem rather than the answer. Required simulator speedup spans $65\times\text{--}287,046\times$, but the ratio alone cannot identify a QPU target. The native solver sets a workload-specific deadline; the circuit path adds a workload-specific quality gap and repetition structure; and the physical stack can move its bottleneck after one resource improves. QArchGauge converts this paired boundary into five explicit outputs: whether the output is valid, which resource to improve first, how much it must improve, which resource becomes next, and the maximum benefit of improving each resource.

The evidence must also separate verifiability from width. Exact 4–20-qubit records close an auditable loop among a native answer, executable circuit, and natural-unit quality. Such scales remain common in end-to-end QML and chemistry studies, which use two-to-seven-qubit learning circuits or reduced five-to-fourteen-qubit active spaces [7]–[9]. Wider 100–

TABLE I: Closest approaches to an application-level architecture boundary. ND: measured native deadline; QT: direct quality target; AL: repeated application loop; HP: physical projection; MU: component marginal utility. $\circ$ denotes partial support.

Study class	Representative	ND	QT	AL	HP	MU
App. benchmark	SuperMarQ, QED-C [2], [3]		✓	◦		
Classical/QC	QUARK [12]	◦	✓	✓		
Full-stack FT	Beverland et al. [13]	◦	◦	✓		◦
Trace/control	TraceQ, d’ArQ, Pinball [14]–[16]			✓	✓	✓
QPU component	BOSS, LSQCA, S-SYNC [17]–[19]			◦	✓	✓
HPC/QC runtime	QFw, ScaleQsim [5], [21]	◦	✓			
QArchGauge	Paired quality-first inversion	✓	✓	✓	✓	✓

127-qubit Ising and shallow-QAOA demonstrations establish processor capability, not same-input advantage over native HPC [10], [11]. We use 36–40-qubit distributed runs only to validate simulator capacity and use deployment-facing CIFAR-10, all-electron CCSD(T), MaxCut, and matrix-free Krylov paths to tighten the native boundary. Neither qubit count nor GPU count is treated as advantage evidence.

Table I states the novelty boundary. SuperMarQ and QED-C provide application benchmarks, circuit features, quality, and time-to-solution [2], [3]. QUARK already compares classical and quantum application implementations [12]. Beverland et al. compile scaled applications into fault-tolerant requirements and are the closest resource-estimation neighbor [13]. TraceQ reconstructs surface-code dataflow, while d’ArQ and Pinball improve causality-aware control and cryogenic pre-decoding [14]–[16]. BOSS, LSQCA, S-SYNC, and Cyclone optimize movement, memory, or QCCD organization [17]–[20]. QFw and scalable simulators provide portable hybrid execution and evidence generation [5], [21]. These systems optimize or estimate a circuit-side mechanism. They do not jointly bind direct application quality, a measured same-input native deadline, case-level reliability, and first-to-next component marginal utility.

Key Idea. QArchGauge uses leadership-class HPC as an empirical oracle for deciding which QPU resource delivers the next unit of application-level progress. One record preserves the input, native deadline, output tolerance, circuit demand, finite-shot evidence, dependency schedule, and provenance. The inversion first rejects records without a same-record full-loop quality pass. It then derives case-level distance and rotation precision, prices factory, logical, decoder, host, and space terms under explicit overlap bounds, and recomputes the complete model after each resource improves. Published mechanisms enter only by replacing matched events; their reported average speedups are never borrowed as universal multipliers.

This paper makes three contributions. First, it defines a

quality-first paired boundary that distinguishes measured, projected, conditional, and deployment evidence and prevents a quality-failing family from receiving an application-level hardware target. Second, it introduces a component-preserving critical-path inversion that combines workload-valid ready widths, finite shots, application failure budgets, code distance, rotation synthesis, factory footprint, decoder reaction, joint design coverage, and the maximum end-to-end gain from each component. Third, an auditable Perlmutter campaign produces 3,552 records across 222 structural configurations and 16 seeds per configuration, plus direct quality-cost and deployment-facing frontiers. Only 12 finite-shot Sim. records close the complete application contract. Their strict surface-code envelope remains T-state-supply-limited until 2.52–2.75 million parallel factories, after which shot parallelism or logical-gate latency dominates. A matched LSQCA replacement further shows how a published architecture can be rejected for this phase: it saves no core cells at 4–7 logical qubits and matched movement eliminates all runtime-favorable cases.

II. BACKGROUND

A. Application Boundary and Evidence Scale

A quantum application wraps a classical input in a circuit and returns samples, observables, kernel entries, or energies to a classical workflow. The kernel is only one stage. Encoding, repeated execution, measurement aggregation, optimizer/search iterations, and postprocessing remain part of application time and quality.

The four families exercise different paths. A quantum neural network or variational classifier (QNN/VQC) trains a parameterized feature circuit through classical feedback; QKernel repeatedly evaluates feature-map circuits to construct a kernel matrix; QFeature executes a fixed circuit and trains a classical head. Chem. uses a VQE ansatz and grouped Hamiltonian measurements. Opt. uses QAOA phase/mixer circuits and sampled objectives. Sim. maps Hamiltonian terms to Trotterized evolution. Their native counterparts are classifiers, eigensolvers, exact or heuristic optimization, and Krylov time evolution. Each pair shares an input and reports quality in accuracy, Hartree, objective ratio, or observable error.

A case crosses the application boundary only when

$$T_{\text{qhw}} < T_{\text{native}} \quad \text{and} \quad \Delta Q \leq \epsilon_w,$$

where ϵ_w is the workload’s natural-unit tolerance. More shots may narrow estimator error, while a deeper ansatz, search, or Trotter product may reduce algorithmic error; both increase hardware demand. Quality recovery is therefore measured jointly with cost rather than represented by a free scalar.

Evidence scale and circuit width are distinct. Controlled 4–20-qubit records permit exact native answers and natural-unit quality closure. Distributed 36–40-qubit state-vector runs establish HPC simulation capacity only. Deployment-facing native workloads tighten T_{native} but do not inherit the controlled circuit’s quality. A projected point becomes an application-level hardware target only when finite-shot output, the com-

plete application loop, dependency schedule, reliability, and physical model attach to the same record.

B. Reliability-Constrained Critical Path

Surface-code execution couples reliability, time, and space. We use the QDK-aligned logical-error envelope

$$p_L(d) = 0.03 (p/0.01)^{\lfloor (d+1)/2 \rfloor},$$

and choose the smallest odd distance whose logical-volume union bound fits the allocated application failure budget [22]. A strict contract requires the complete estimator’s logical, synthesis, and distillation failures to fit 1% or 0.1%. A separately reported estimator-tolerant contract permits corrupted shots only when direct finite-shot output provides enough measured headroom; it does not assume that every shot must be error-free.

Arbitrary rotations add a second reliability constraint. The leading Ross–Selinger synthesis cost is $3 \log_2(1/\epsilon_{\text{rot}})$ T states, where the per-rotation precision budget follows from the number of rotations in the application loop [23]. A fixed 16- or 32-T assumption is invalid whenever it exceeds that budget. Factory output is then priced with LSQCA’s 176-cell surface-code factory producing one state every 15 code beats, while QDK’s Litinski protocols supply the corresponding physical-qubit footprint [18], [22]. These are separate throughput and space envelopes; they are not multiplied together as independent speedups.

For logical work T_{log} , decoding/reaction T_{dec} , and factory supply T_{fac} , unavailable contention is bracketed by

$$T_{\text{core}}(\rho) = \max_i T_i + \rho \left(\sum_i T_i - \max_i T_i \right), \quad \rho \in [0, 1].$$

Decoder service and the communication/control path remain separate because reaction time can require correction storage and additional protected memory [24]. Host-visible feedback is serial only when the recorded loop requires it. Likewise, useful shot lanes are capped by dependency-ready repetitions, not physical lane count alone. A component design such as LSQCA, BOSS, S-SYNC, Pinball, or Cyclone can replace only matched core-cell, movement, decoder, or topology events [16]–[20]. This replacement rule is what lets an application boundary identify when a published mechanism is first-order and when it is not.

III. DESIGN

QArchGauge turns an application comparison into a replaceable architecture contract. Its invariant is that the input, native deadline, quality target, application loop, and provenance do not change when one QPU mechanism is improved.

A. Paired Evidence Record

Figure 2 begins with one concrete input/configuration tuple. Both paths receive the same instance identifier, seed, transform, molecule, graph, or Hamiltonian and the same natural-unit quality metric. The native path records every valid non-quantum candidate and selects the fastest one satisfying

Quality-first critical-path inversion

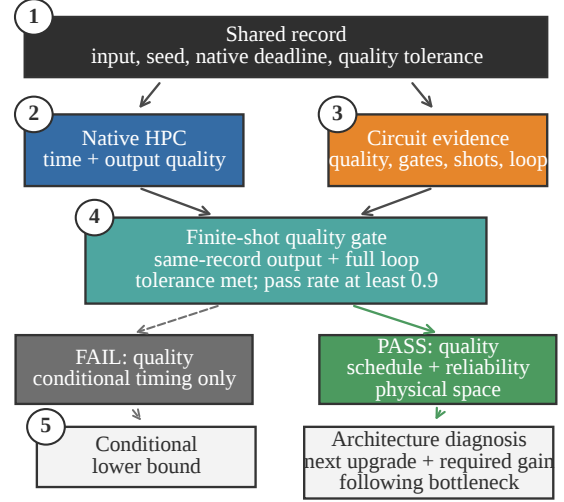


Fig. 2: QArchGauge’s two-stage inversion. Solid arrows carry eligible evidence; the dashed branch retains quality failures as conditional lower bounds. Only a passing record receives a physical diagnosis: the best next upgrade, its required magnitude, the following bottleneck, and the maximum gain from each resource.

the native target. The circuit path records construction and simulator time, output quality, gates, measurement groups, shots, evaluations, and raw source. Failed candidates remain visible. A stronger native method simply replaces T_{native} and moves the boundary outward.

Simulator execution recovers the exact circuit output and demand; it never sets a QPU latency. The record explicitly tags values as measured, modeled, stress, or future replacement. Controlled records provide paired quality; deployment-facing records strengthen only the native frontier; large distributed simulations provide only capacity evidence. This typing prevents a 256-GPU throughput result or a 40-qubit state-vector run from becoming a QPU-scaling claim.

Records are not independent applications. A structural configuration fixes the workload instance and algorithm settings; seeds and repeat rounds provide independent observations within that configuration. Family statistics therefore report records, structural configurations, independent instance/seed pairs, and repeats separately. Claim intervals use a hierarchical bootstrap over independent instance/seed units, and workload-balanced aggregates give each family equal weight.

B. Quality-First Physical Inversion

The first stage partitions every record into a noiseless quality pass, a failure with a measured recovery ladder, or a failure without recovery evidence. Representative records are then replayed at 10^3 , 10^4 , and 10^5 shots with independent sampling seeds. Hardware-target eligibility requires at least 0.9 replicate pass probability, a measurement contract for the same record, and coverage of the full application loop. A fixed selected

TABLE II: Inversion contract and replacement boundary.

Term	Authoritative evidence	Allowed replacement
Quality	Natural-unit output; 12 finite-shot replicates	Measured richer encoding, depth, search, or mitigation output
Ready width	Compiled group or source-audited static loop	Compatible dependency trace; no invented queue
Reliability	1%/0.1% application budget; QDK-aligned d	QEC/logical-error model with equal or stricter scope
Rotations	Emitted/reconstructed angles; precision-derived T count	Compatible synthesis or native rotation, including its success cost
Factory/space	LSQCA throughput; QDK protocol footprint	Matched supply/layout events, not a reported mean speedup
Decoder/control	Service plus 7.8- μ s communication path	Measured reaction, bandwidth, storage, and tail trace

VQE or QAOA parameter has useful output-quality evidence but does not close its outer optimizer. Such records and all other quality failures report *algorithm/representation first*; their physical timing is retained only as a conditional execution lower bound.

For an eligible record, source or compiled schedules expose three ready-width bounds: all predeclared work ready, one measurement-group wave ready, and one evaluation ready. If P_s shot lanes exist but only P_{ready} repetitions can issue, $P_s^{\text{eff}} = \min(P_s, P_{\text{ready}})$. We populate P_{ready} from compiled Chem. groups, source-audited static ML/Chem./Opt. loops, or the single Sim. circuit. No adaptive optimizer or mid-circuit branch is invented.

The strict reliability contract splits an application failure budget ϵ_a equally among logical, rotation-synthesis, and distillation faults. Routed logical locations and all finite-shot repetitions select d ; the number and precision of arbitrary rotations select T states per rotation and a compatible QDK factory protocol. For distance d , physical syndrome round τ_s , N_T T states, F factories, and P_s^{eff} lanes, the principal terms are

$$\begin{aligned} T_{\text{log}} &= W_{\text{cyc}} d \tau_s / P_s^{\text{eff}}, \\ T_{\text{fac}} &= 15 N_T d \tau_s / F, \\ T_{\text{dec}} &= N_{\text{wave}} (L_{\text{service}} + L_{\text{comm}}). \end{aligned}$$

The model adds host-visible rounds and recomputes the overlap bracket from Section II. Physical output includes data qubits, selected factory-protocol qubits, LSQCA's 176-cell supply envelope, syndrome bandwidth, and reaction-time correction storage. Factory-internal syndrome traffic, rotation anticommutation depth, and uncompiled adaptive traces remain explicit unsupported flags.

C. Joint Target and Mechanism Inversion

For each eligible record and resource j , QArchGauge recomputes the complete model after a $10\times$ improvement and after making that resource non-limiting:

$$G_j^{10} = T/T_j^{10}, \quad C_j = T/T_j^{\infty}.$$

G_j^{10} determines the first useful marginal target; C_j is its Amdahl-style removal ceiling. The next target is computed only after removing the first and re-evaluating overlap, saturation, and feedback. This avoids ranking components by isolated bars that cannot execute concurrently. A record already satisfying quality, reliability, and runtime is labeled advantage-reached rather than assigned an unnecessary target.

One-factor results are insufficient when factory supply, useful lanes, logical cycle, decoder reaction, native deadlines, and overlap interact. The joint design therefore uses a deterministic, marginally balanced space-filling set over dependency mode, $0.5/1/2\times$ tolerance, 10^3 – 10^5 direct shots and useful lanes, fixed or precision-derived T counts, 1%/0.1% failure budgets, $10^{-3}/10^{-4}$ physical error, 1–50,000 $\times$ factory supply, 0.1/0.4- μ s syndrome rounds, 1/5/63- μ s decoder service, $0.5/1/2\times$ native deadlines, and $\rho = 0/0.5/1$. These points measure design coverage, not probabilities of future technologies. A phase cell reports the first-target mode, agreement count, runtime-pass count, successor, and removal ceiling.

Published mechanisms use the same invariant. Our LSQCA replacement reconstructs each workload's ordered logical-operand events, applies a two-entry computation-register policy, and charges point-SAM's $7\sqrt{n}$ -beat load and reverse store. Only core cells and matched movement change; quality, reliability, rotations, factory demand, and native time remain fixed. A separate BOSS-compatible envelope consumes direct Chem. and QAOA interaction graphs to bound shuttle, SWAP, recooling, and readout under locality-first and congestion-stressed policies. It is not presented as the full BOSS compiler or as a universal QCCD multiplier.

D. Typed Architecture Decision

The output is not a scalar speedup. For record r , QArchGauge emits

$$\mathcal{D}(r) = \langle e, j_1, x^*, j_2, \{C_j\}, \pi \rangle,$$

where e is the evidence state, $j_1 = \arg \max_j G_j^{10}$ is the first compatible resource, x^* is its first crossover, j_2 is the target after that crossover, C_j is each removal ceiling, and π resolves every term to measured, modeled, stress, or replacement provenance. The evidence state is one of quality-blocked, loop-blocked, FT-invalid, hardware-target, or advantage-reached. A blocked record returns algorithm/representation or loop closure as j_1 and leaves physical x^* undefined; it cannot inherit a component target from another record in the same family.

For a scalable resource, x^* is the smallest tested or solved multiplier at which its first-target identity changes or complete runtime reaches native parity. The successor is then recomputed from the new operating point, not copied from the original component ranking. At family level we report the count and agreement fraction of record-level decisions. Agreement is deterministic design coverage, not a probability that a future machine will occupy that point. This typed output is what separates an architecture decision from a simulator benchmark: it states what is blocked now, what change is

useful first, how much change is needed, and what becomes limiting next.

A proposed component is evaluated by replacing only the events it owns and regenerating $\mathcal{D}(r)$. The proposal earns application-level priority only if it changes j_1 , reduces x^* , raises a runtime-and-quality pass, or exposes a new successor without violating reliability or adding unmatched costs. A large local speedup with unchanged $\mathcal{D}(r)$ is reported as low application-level leverage. This rule admits future measured QPU traces, native rotations, decoders, layouts, and HPC-QPU runtimes without changing the paired application contract.

Implementation and audits. Perlmutter records capture Python/CUDA/library versions, Slurm provenance, seeds, failures, and raw timing. Qiskit compiles routing and commuting groups; QDK and Qualtran cross-check selected logical circuits. Independent audits verify quality eligibility, schedule provenance, reliability/space, statistical units, joint phase coverage, replacement invariants, measured-versus-modeled labels, and paper claims. Every retained result resolves to JSON/CSV evidence and a deterministic regeneration script.

IV. EVALUATION

The evaluation follows the inversion order in Figure 2. We first establish a deployment-facing native frontier, then determine which circuit records satisfy application quality, and only then derive logical and physical architecture targets. Perlmutter is the evidence engine throughout; its GPU time is never substituted for QPU time.

A. Evaluation Setup and Contracts

Platform and evidence units. Experiments run on NERSC Perlmutter GPU nodes, each with four NVIDIA A100 GPUs. The controlled corpus contains 3,552 records: 2,048 ML, 224 Chem., 768 Opt., and 512 Sim. records. These records cover 222 structural configurations and 16 distinct seeds per configuration. A structural configuration fixes the input and algorithm settings; a seed within that configuration is an independent observation, not a repeated timing trial. We report hierarchical-bootstrap 95% intervals over configurations and seeds using 2,000 resamples, and macro-averages weight the four families equally. Warmup-separated stability trials and all raw failures are stored separately.

This distinction changes the reported uncertainty. The workload-balanced noiseless quality-pass fraction is 17.9% with a 95% interval of 11.2–24.8%, rather than a raw-record average dominated by ML. The geometric mean of the four family-median simulator/native ratios is $19,329 \times$ ($18,108$ – $26,829 \times$). Twelve warmup-separated timing trials, three per family, have a maximum quantum-runtime coefficient of variation of 4.0%. Capacity probes and deterministic design scenarios are reported as bounds rather than assigned sampling confidence.

Table III defines the paired application contract. A quality pass requires the quantum path to remain within the listed natural-unit tolerance of the native output. The main corpus is exact/noiseless and exposes representation or approximation

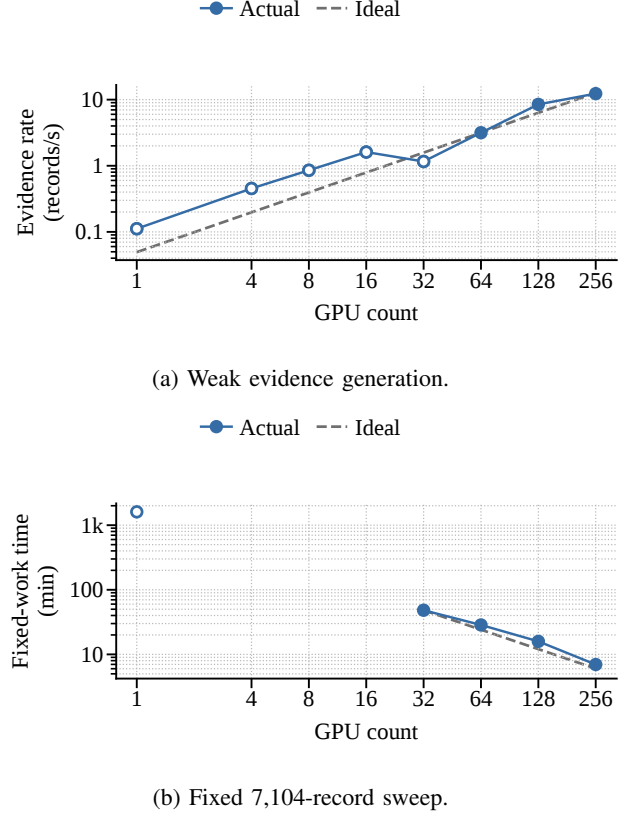


Fig. 3: Perlmutter evidence-generation scaling. Actual and ideal lines compare completed corpus runs; the hollow 1-GPU point is a split-array anchor. Timeout-censored 4/8/16-GPU attempts are not plotted as fixed-work completions.

TABLE III: Same-input application contracts. Deployment-facing native cases strengthen the frontier but do not alter controlled-corpus quality.

Family	Native path	Circuit path and quality gate
ML	scikit classifiers; matched Pool-108; A100 ResNet-18	QKernel, QNN/VQC, or QFeature; accuracy loss ≤ 0.02
Chem.	dense/sparse eigensolver; FCI/CASCI; CCSD(T)	VQE-style preparation and Pauli energy; error ≤ 0.01
Opt.	exact, local search, annealing, or MILP	QAOA sampling; approximation-ratio loss ≤ 0.02
Sim.	dense or matrix-free Krylov evolution	Trotter circuit and terminal observable; error ≤ 0.01

error. A separate direct experiment selects 68 source records and samples each at 10^3 , 10^4 , and 10^5 shots with 12 independent replicates. Hardware eligibility additionally requires at least 0.9 pass probability, a measurement trace for the same record, and coverage of the complete application loop.

Scale and provenance. Figure 3 verifies that Perlmutter increases the rate at which paired application records are built. With about 28 records per GPU, aggregate throughput reaches 12.3 records/s on 256 GPUs. For the completed direct 7,104-record runs, elapsed time falls from 2,894 s on 32 GPUs to 418 s on 256 GPUs, a $6.9 \times$ speedup. The hollow 1-GPU point

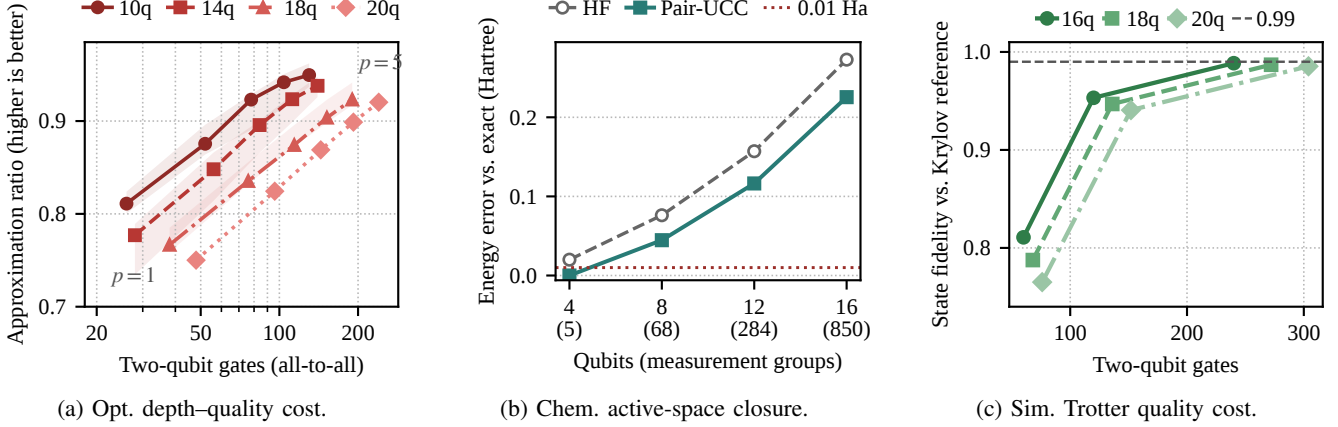


Fig. 4: Quality recovery consumes circuit work. Opt. depth improves approximation ratio while multiplying routed two-qubit gates; Chem. measurement groups grow while larger active spaces remain outside chemical accuracy; Sim. state fidelity improves with Trotter depth.

is a split-array anchor. Separate 4/8/16-GPU attempts reached 4,298/5,794/7,092 records before their 8/5/3-hour limits; they are censored and never extrapolated into measured completion times. This is case-level parallelism, not distributed execution of one circuit.

We separate semantic closure from state-vector capacity. The controlled applications remain at 4–20 qubits so that their native answer, circuit output, and quality gap can be checked exactly. Separate completed probes hold a 36-qubit, 0.5-TiB state on 64 GPUs; a 38-qubit, 2.0-TiB state on 128 GPUs; and a 40-qubit, 8.0-TiB state on 256 GPUs. The corresponding shards are 8, 16, and 32 GiB/GPU. Median local-gate time remains within 0.614–0.807 s as aggregate state grows by $16\times$, while the 38-qubit maximum exposes a 29.888-s straggler. Each rank transforms only its local shard, so these runs establish storage reach and process coverage, not a complete cross-rank application or QPU scaling.

The physical analysis instead consumes gates, angles, shots, dependency-ready work, quality, and native deadlines from the paired records. All 224 Chem. records have compiled measurement groups, 3,040 ML/Chem./Opt. records have source-audited static loops, and all 512 Sim. records have one terminal circuit. No adaptive optimizer or mid-circuit trace is inferred. Keeping the capacity probes separate prevents a large stored vector from being mistaken for large-scale application-quality closure.

B. Deployment-Facing Native Frontier

The controlled suite is intentionally small enough to close input, circuit, and exact quality. We therefore add deployment-facing native paths before projecting hardware. A stronger native implementation only shortens T_{native} and moves the break-even boundary outward; it cannot manufacture a favorable quantum result.

ML representation. Figure 5 compares three paths on the same CIFAR-10 50,000/10,000 split. A one-A100 ResNet-18 trained for 10 epochs with AMP/TF32 takes 56.5 s and

reaches 81.85% accuracy. QFeature uploads 4,096 amplitudes to 12 qubits, applies three entangling layers, extracts 108 observables, and trains a ridge head; it takes 115.16 s and reaches 33.60%. Pool-108 uses the same feature count and ridge head but fixed spatial pooling, taking 2.45 s and reaching 40.28%. Thus the circuit representation is $46.97\times$ slower and 6.68 points less accurate than its matched control before QEC, shots, or generic state preparation. Faster logical gates cannot close this representation gap.

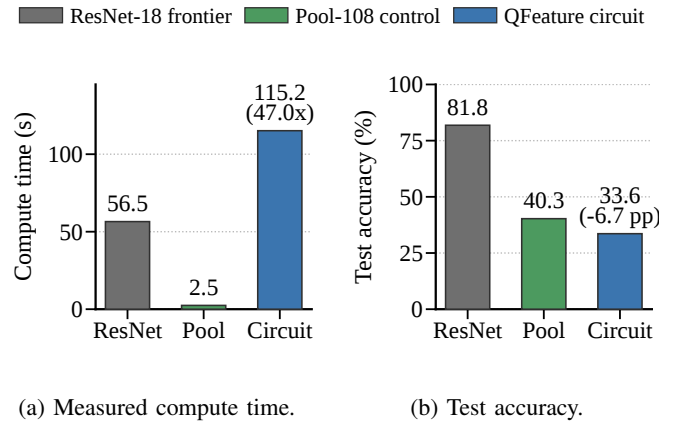


Fig. 5: Matched CIFAR-10 outcomes. Pool-108 and QFeature expose 108 features to the same ridge head; direct annotations show that QFeature is $47.0\times$ slower and 6.7 points less accurate than this matched control. ResNet-18 provides the deployment-facing native quality frontier.

Quality costs beyond ML. Figure 4 measures the work needed to recover quality rather than assuming that recovery is free. For weighted MaxCut, increasing directly optimized QAOA from $p = 1$ to 5 raises the median 10/14/18-qubit approximation ratio from 0.785 to 0.938, but raises all-to-all 2Q work from 28 to 140 gates; line routing expands it by $2.15\text{--}2.56\times$. The 20-qubit $p = 5$ case reaches 0.920 with 240 all-to-all or 898 line-routed 2Q gates. Native 50–100-node local

search and annealing reach median 0.998 and 1.000 of the best observed objective.

For Chem., nested 4/8/12/16-qubit H_8 active spaces compare pair-UCC with exact FCI. Only the 4-qubit space closes 0.01 Ha; best errors are below 10^{-8} , 0.04465, 0.11637, and 0.22547 Ha while qubit-wise-commuting groups grow from 5 to 850. Deployment N_2/H_2O CCSD(T) instances map to 120/184 spin-orbital qubits, but do not have a matched large-space VQE-quality closure and are not assigned a physical target. For Sim., 20-qubit Trotter fidelity rises from 0.765 to 0.985 at 4/8/16 steps while 2Q gates rise from 76 to 304; the native matrix-free Krylov reference takes 0.56–20.56 s at 16–20 qubits with norm error below 6×10^{-14} . These results identify representation, ansatz/search, and approximation order as first-class architecture inputs rather than NISQ-noise artifacts.

Figure 6 applies an adversarial native stress using the A100 tensor-core/HBM peaks and a 10- μ s launch floor under the Roofline method [25]. It shortens the implemented native deadline by a median $209 \times / 17 \times / 2.4 \times / 208 \times$ for ML/Chem./Opt./Sim. The stress is an unattainable lower bound, not a measured implementation or a SOTA claim. Its role is to test direction: replacing any executed native path with a more optimized one can only make the quantum boundary harder. We retain the measured deadline in subsequent projections, so their runtime targets remain optimistic with respect to this stress.

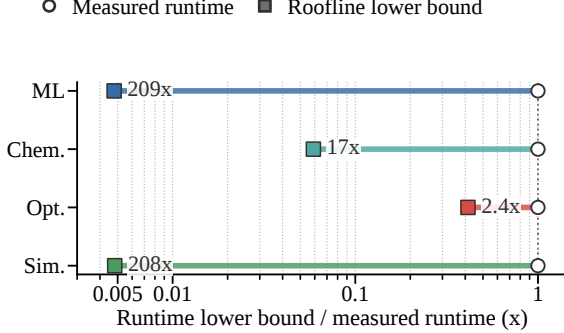


Fig. 6: Adversarial native-runtime stress. Circles normalize each measured family median to 1; squares show the peak/HBM lower bound. Moving left gives the circuit path less time to beat native HPC; the span is not measured application speedup.

Observation 1: A stronger native implementation moves the advantage boundary outward, while better circuit quality adds gates, measurements, or search work. QPU targets must therefore use the paired native deadline and charge the measured cost of quality recovery.

C. Quality-Qualified Application Boundary

Hardware speed matters only after the circuit output is useful. Figure 7 therefore applies the quality gate before any hardware projection. In the full noiseless corpus, no ML or Opt. record passes; 48/224 Chem. and 256/512 Sim. records

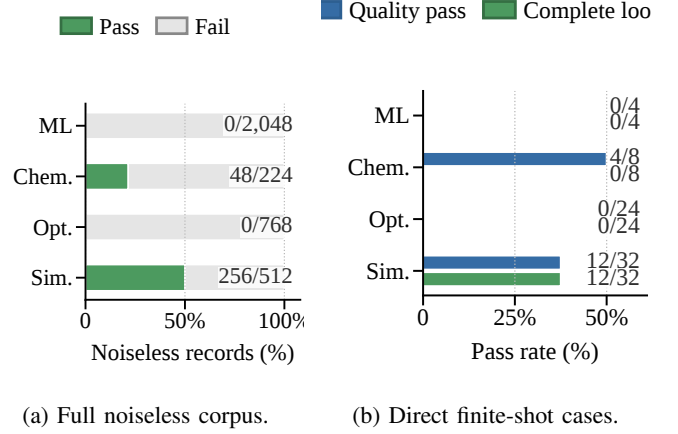


Fig. 7: Quality eligibility precedes hardware targeting. The 3,552-record noiseless corpus yields 304 quality passes. A separate 68-record direct audit at 10^4 shots yields 16 quality passes, but only 12 Sim. records cover the complete same-record application loop and become physical targets. Counts beside every bar retain the panel-specific denominator.

pass. The 95% hierarchical-bootstrap intervals are [0,0], [0,0], [0,0.429], and [0.344,0.688], respectively. Halving or doubling each tolerance preserves this ordering.

At 10^4 shots, ML passes 0/4, Chem. 4/8, Opt. 0/24, and Sim. 12/32. The Chem. passes sample one fixed parameter set and do not include VQE optimization. ML changes to a measurable Z/ZZ feature map but still misses the accuracy bound. Opt. samples fixed QAOA parameters and omits the outer search. These are algorithm and loop gaps, not NISQ noise: the source corpus is noiseless. Only the 12 Sim. cases at the declared 10^4 -shot contract are same-record, full-loop, quality-qualified hardware targets. Physical numbers for every other family are explicitly conditional lower bounds. At 10^3 shots no case passes; at 10^5 , Chem. remains 4/8 and Sim. rises to 16/32.

The split follows application-loop structure. ML and Opt. lose quality before hardware timing begins, while the passing Chem. outputs still omit the optimizer that would repeatedly request energies. Sim. has one terminal observable and no hybrid outer loop, so its passing records can close the full application contract. The first architecture action is therefore family-specific: improve representation or search for ML/Opt., close the measured VQE loop for Chem., and only then optimize QPU resources for eligible Sim.

Observation 2: Faster gates cannot rescue an output that fails the application contract. ML, Chem., and Opt. first need better representations, ansatzes, or searches; only the 12 full-loop Sim. records support physical hardware targets.

D. Trace-Aware Logical Lower Bound

The next test gives the QPU free magic states and asks whether logical execution alone can beat native HPC. It removes simulator wall time and factory latency, but keeps direct 10^4 shots, case-level reliability, logical work, useful-

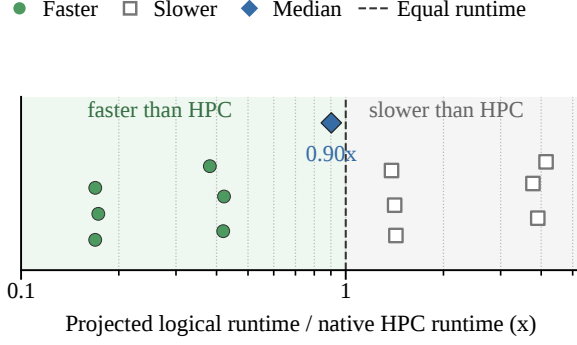


Fig. 8: Logical-runtime lower bound with free T-state generation at 10^4 shots. Each point is one eligible Sim. record; the dashed line marks equal native-HPC runtime and the diamond is the median.

lane limits, decoder reaction, and host-visible rounds. Figure 8 shows only the 12 Sim. cases that passed the full-loop gate in Figure 7.

The eligible records span $0.170\text{--}3.891\times$ native time from the 10th to 90th percentile. Their median is $0.903\times$, and 6/12 beat native time. The schedule variants coincide because each record has one terminal circuit. For context, the same optimistic model gives conditional medians of $90.6\times$ for ML, $526.2\times$ for group-ready Chem., and $3,653.9\times$ for Opt.; these are not hardware targets because they failed Figure 7. Removing Perlmutter simulation cost therefore does not guarantee advantage. Finite-shot logical work already blocks half of the eligible records.

This split is not a simulator constant. Each ratio combines that record’s measured native time with its own logical work and direct shot count. The same workload family therefore falls on both sides of parity. An architecture must reduce useful logical work or execute independent shots concurrently without weakening the quality contract; a single average gate-speed claim cannot predict which records cross the boundary.

The lower bound is also an early rejection test for component proposals. All 12 records use only 4–7 logical qubits, yet their ratios span both sides of native parity; qubit count alone does not determine the result. A mechanism that changes T-state factories, movement, or decoder service cannot rescue an open-square record that is slower than native HPC, because those costs are already free or absent from this test. Such a record first needs a shorter logical schedule, more dependency-ready shot concurrency, or fewer shots at the same measured quality. For a filled-circle record that is already faster, those logical conditions are sufficient and the physical stack becomes the next question. The test therefore prevents a downstream component gain from being credited before the record reaches that component.

Observation 3: Even with free T states, only 6/12 eligible records beat native HPC. Their first useful target is lower logical and shot work, while the other six already have enough logical speed and expose the physical factory cost next.

E. Reliability-Constrained Physical Diagnosis

Figure 9 restores arbitrary-rotation synthesis and distillation for the 12 eligible records. Under a 1% application-failure budget and physical error 10^{-3} , the all-shots contract, aligned with QDK, selects distance 13–15 and 79–86 T states per arbitrary rotation. A sensitivity that allows a bounded number of failed shots selects distance 7–9 and 49–58 T states. Panel (a) exposes both resource settings rather than hiding them behind one global distance or T count. The compatible all-shots protocol uses 2,070 physical qubits per factory output path [23], [26].

The all-shots contract requires 2.52–2.75 million factories operating in parallel, equivalent to $39,351\text{--}42,961\times$ the 64-factory baseline. Their protocol footprint is 5.21–5.69 billion factory qubits. This is a supply-equivalent target, not a floor-plan. It is large because every arbitrary rotation is repeated over 10^4 shots while the reliability budget tightens both T-state error and code distance. Allowing a bounded failed-shot tail reduces the target to 1.53–1.84 million factories, distance 7–9, and 49–58 T states per rotation, but does not change the first bottleneck. Decoder reaction combines $7.8\ \mu\text{s}$ communication with $1/5/63\ \mu\text{s}$ service envelopes [24]; at $5\ \mu\text{s}$, median syndrome bandwidth is 3.09 Gb/s. Control becomes important later, after factory supply is no longer dominant.

Space gives the same diagnosis. Across the 12 eligible records, protected data occupies 1,352–3,150 physical qubits, while the compatible QDK factory protocols occupy 5.21–5.69 billion qubits at crossover. Mapping the same supply into LSQCA’s 176-cell density envelope gives a separate 150–218-billion-qubit upper envelope; the two models are reported separately, not multiplied. At the $5\text{-}\mu\text{s}$ decoder point, reaction-time correction storage alone spans 2.56–3.71 billion qubits and the data-block syndrome stream spans 1.69–3.94 Gb/s. These values do not propose such a machine. They show that synthesized-rotation supply and its protected state dominate the data block before decoder or layout improvements can become first-order.

Observation 4: Fault tolerance reverses the favorable result with free T-state generation. The first physical target is not a slightly faster gate, but four to five orders of magnitude more useful T-state throughput; relaxed reliability reduces its size but not its identity.

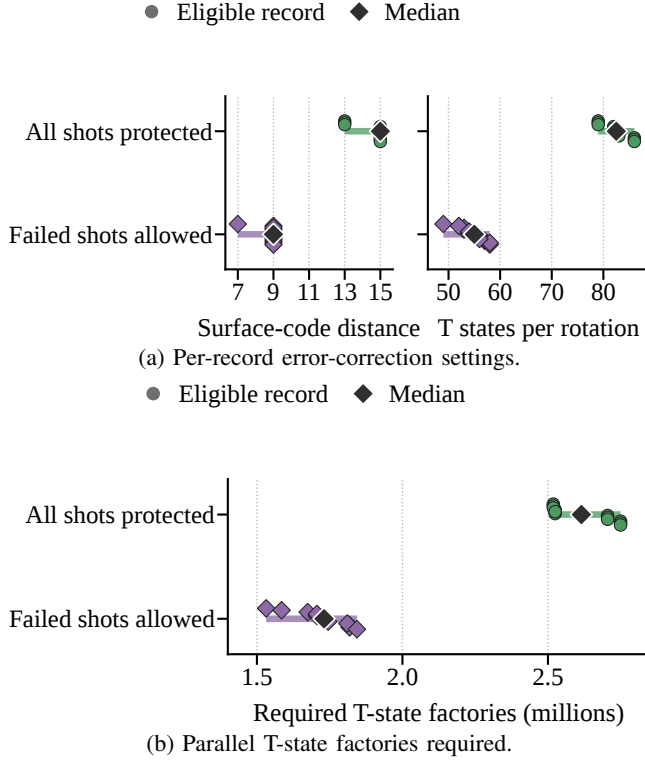


Fig. 9: Reliability choices set physical cost. Points are the 12 eligible records and diamonds mark medians. (a) Protecting every shot requires larger codes and more T states per rotation than allowing a bounded failed-shot tail. (b) The two contracts require 1.53–2.75 million parallel T-state factories before T-state generation stops dominating. Failed-shot tolerance is a sensitivity, not the default.

F. Joint Co-Design Sensitivity

Improving one resource moves the bottleneck to another. We evaluate 384 deterministic scenarios across quality, shots, useful lanes, T count, reliability, physical error, factory supply, logical cycle, decoder service, native time, dependencies, and overlap. Figure 10 keeps the all-shots contract and shows which $10\times$ resource upgrade reduces runtime most for each of the 12 eligible records. Repeated cells with identical outcomes are collapsed.

Eligibility is recomputed inside each scenario rather than held at 12 by construction: changing shots or tolerance can move a record across the quality gate. Likewise, fixed 16- or 32-T rotation scenarios that violate the precision-derived synthesis budget are retained as invalid sensitivities but cannot support a physical target. The plotted slice fixes the declared 10^4 -shot quality contract, reliability-derived T count, 1% application budget, and 10^{-3} physical error so that every transition is executable under one consistent contract.

The complete space-filling design contains 128 reliability-derived-T points in addition to invalid fixed-T stresses. Forty-one have no valid eligible modal target and remain unclassified rather than being forced into a hardware category. Of the remaining 87, 60 are factory-first, 13 shot-first, four logical-cycle-first, and ten already reach advantage. Supply explains

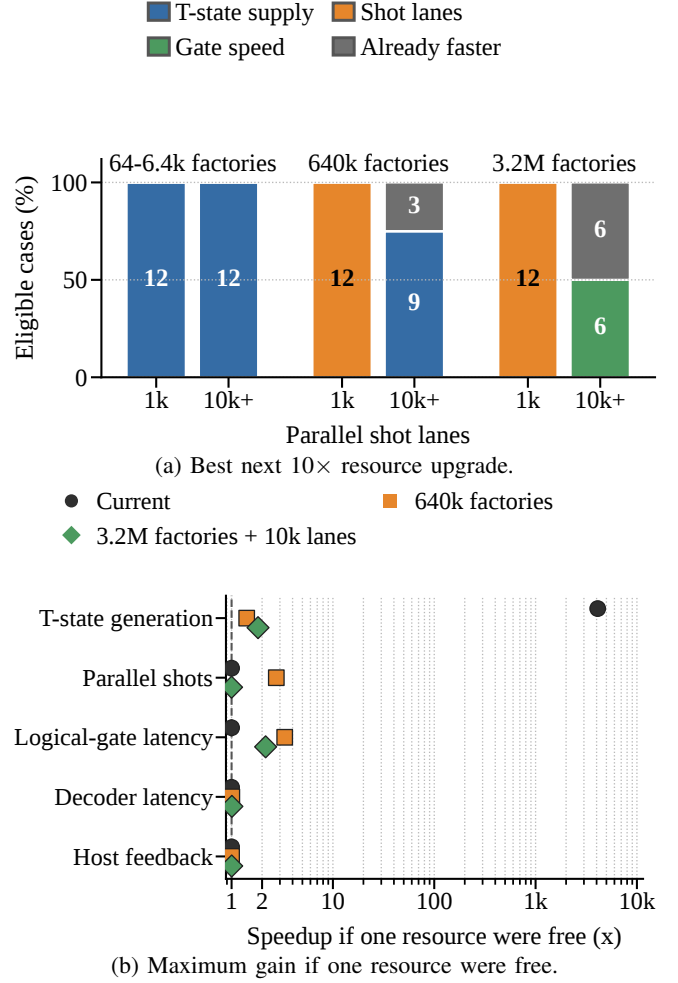


Fig. 10: How the next useful upgrade changes. (a) Each colored segment counts records for which a $10\times$ improvement to that resource reduces runtime most; “already faster” means projected runtime is below native HPC. Group headings give absolute T-state factory counts and ticks give parallel shot lanes. (b) Each marker is the median speedup if one resource cost becomes free; $1\times$ means no useful leverage.

the transition without assigning scenario probabilities: at 1–100 $\times$ factory supply, 44/45 valid modal decisions are factory-first; at 10k $\times$, the valid modes split into 12 factory, eight shot, one cycle, and two advantage; at 50k $\times$, they split into four, five, three, and seven. Figure 10 isolates the declared contract so the causal handoff is not hidden by quality and FT-invalid points.

At 1–100 $\times$ factory supply, factory supply is first for every lane count. At $10^4\times$, 1,000 lanes makes shot parallelism first; with at least 10,000 lanes, three records reach advantage and nine remain factory-limited. At $5 \times 10^4\times$, 1,000 lanes is still shot-limited; at least 10,000 lanes leaves six advantaged and six logical-cycle-limited records. Thus the useful order is factory throughput, shot lanes, and then logical-cycle latency. Decoder and host latency do not become first in this slice.

Panel (b) separates the next $10\times$ action from the maximum

● Record □ Q1--Q3 — Median --- Equal runtime

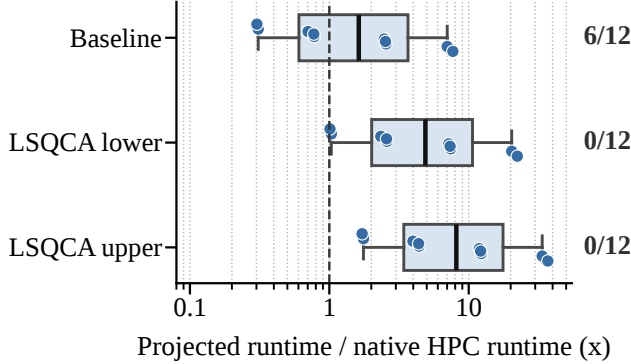


Fig. 11: Matched-event LSQCA replacement. Points are the 12 eligible records; boxes span Q1–Q3, center lines mark medians, whiskers span p10–p90, and the dashed line marks equal native-HPC runtime. Right labels count runtime passes. Only core cells and movement change.

possible gain. At the current point, making T-state generation free gives a median $4,102\times$ speedup, whereas making shot, gate-cycle, decoder, or host cost free gives at most $1.00024\times$. With 640,000 factories and 1,000 lanes, the maximum gains from free shot and gate-cycle cost rise to $2.76\times$ and $3.34\times$; the next $10\times$ test still selects shot lanes first. With 3.2 million factories and 10,000 lanes, gate-cycle cost has the largest remaining gain at $2.17\times$, followed by T-state generation at $1.82\times$, while decoder and host remain below $1.01\times$. The distinction prevents a large asymptotic gain from displacing the resource that gives the next realizable improvement.

Each bar segment is a design decision, not a probability assigned to a future technology. “T-state supply” means that a $10\times$ supply improvement gives the largest compatible runtime reduction for that record; “already faster” means the complete quality-qualified runtime is below native HPC. The transition across columns is therefore causal: adding factories alone eventually exposes serialized shots, and adding useful lanes then exposes logical-gate latency. Collapsing identical settings keeps the figure focused on these transitions rather than repeated parameter combinations.

The framework can also test a published mechanism without borrowing its reported mean speedup. Figure 11 inserts LSQCA point-SAM core cells and matched load/store events into the 5×10^4 -factory, 10,000-lane point while holding all other terms fixed [18]. These 4–7-qubit circuits are below point-SAM’s seven-qubit area break-even, so none gains core area. The lower and upper movement envelopes increase median runtime by $3.12\times$ and $5.23\times$ and remove all six runtime passes. This does not judge LSQCA at its intended scale; it shows that a mechanism must be matched to workload events.

A BOSS-compatible graph envelope also bounds QCCD routing without claiming a device simulation [17]. Chem. adds

■ T-state generation ■ 1Q gates ■ 2Q gates
■ Readout/reuse ■ Move/control

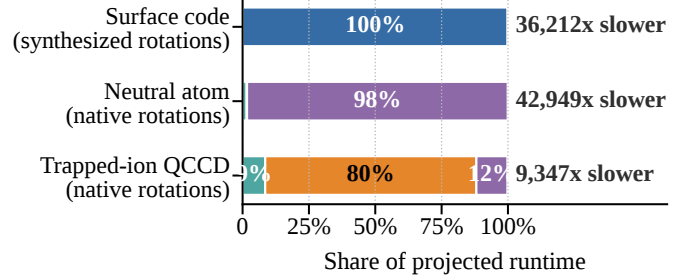


Fig. 12: Where projected time goes under three QPU approaches. Each row is normalized to its own runtime, and the right label reports the absolute slowdown versus native HPC. The surface-code row uses the all-shots contract; native-rotation rows are execution-only counterfactuals calibrated from neutral-atom and QCCD reports [17], [27], without hardware noise or a fault-tolerance claim.

zero locality-first shuttles and at most 24 at the p90 congestion bound; Opt. adds median 7 and 35.5, respectively. Because both families fail the quality gate, these remain conditional routing bounds.

The first target also depends on physical modality. Figure 12 re-prices the same 12 eligible Sim. demands under the all-shots surface-code path and two literature-calibrated native-rotation envelopes. At the measured surface-code point, the median projection is $36,212\times$ slower than native and T-state generation accounts for 99.98% of execution time, consistent with the $4,102\times$ maximum gain when that cost is removed. Replacing synthesized rotations with the neutral-atom envelope moves 98.3% of time to readout/reuse; a $10\times$ readout improvement gives an $8.71\times$ end-to-end gain. The QCCD envelope instead assigns 79.6% to two-qubit gates and 11.8% to readout; a $10\times$ two-qubit-gate improvement gives $3.52\times$. Thus the inversion does not prescribe one universal QPU. It identifies T-state generation for synthesized surface-code execution, readout/reuse for the zoned native-rotation envelope, and entangling gates for the QCCD envelope.

Removing factories is therefore necessary but not sufficient. The neutral-atom and QCCD envelopes remain median $42,949\times$ and $9,347\times$ slower than the paired native deadlines because 10^4 measured shots repeatedly pay readout or gate time. These ratios are not present-device benchmarks: count-based intra-circuit parallelism is optimistic, the neutral-atom control attachment is a lower bound, and the QCCD path prices extra routed gates as recooling rather than running a full transport schedule. Their value is architectural: they show which measured event a specialized accelerator must remove next, while preserving the same finite-shot quality and native deadline.

Resources absent from Figure 10 are also informative. Decoder, host/control, and matched layout work have low

marginal value while rotation supply or shot serialization remains upstream. This is not a claim that they never matter. It says their application-level value must be re-evaluated after the earlier crossover, using the dependency and movement events of the same record. Figure 11 demonstrates the converse: introducing a layout mechanism too early can add enough movement to erase every favorable case.

Observation 5: QArchGauge returns an ordered build decision, not one universal bottleneck. ML and Opt. stop at output quality, and Chem. stops at full-loop closure; none yet justifies a physical-QPU target. For the 12 eligible Sim. records, the all-shots surface-code path first needs less synthesized-rotation demand or $2.52\text{--}2.75$ million parallel T-state factories. Between 640,000 and 3.2 million factories, at least 10,000 parallel shot lanes prevent sampling serialization. After both improve, logical-gate latency becomes first for the remaining six records. Decoder, control, and layout are not first at the measured point, and charging matched LSQCA movement removes all six otherwise favorable cases. The measured build order is therefore algorithm/loop, T-state generation, parallel shots, and logical-gate latency; a component should move forward only when its matched events become first.

V. DISCUSSION AND FUTURE WORK

What should be built first. The result is not a universal request for faster gates. ML and Opt. fail direct quality even without hardware noise, and the passing Chem. outputs do not close the VQE optimizer. Their first target is therefore an encoding, ansatz/search, or mitigation method that preserves the same application output and exposes its complete loop. Soft-information mitigation and postselected rotation preparation illustrate relevant cross-layer directions, but their extra samples, success probability, and control path must enter the paired record [28], [29]. For the 12 eligible Sim. records, the strict FT path is factory-first throughout the $1\text{--}100\times$ region. The multi-million-factory equivalent should be read as a demand mismatch, not a floorplan recommendation: architects must reduce rotation demand, increase compatible state supply, or use a modality that avoids synthesized rotations before decoder or movement improvements can affect application time.

A conditional architecture roadmap. The joint DSE identifies when the roadmap changes. After $10^4\text{--}5 \times 10^4$ factory-supply improvement, insufficient useful lanes make shot parallelism first; when lanes are provisioned, logical-cycle latency becomes first for the remaining cases. TraceQ and d’ArQ can expose dependency and control structure, while Pinball targets predecoding latency [14]–[16]. BOSS, LSQCA, S-SYNC, and Cyclone target movement, storage, or QCCD organization [17]–[20]. These mechanisms are valuable at the operating points that activate their matched events. In our small eligible circuits, LSQCA saves no core cells and movement erases six runtime passes; at larger logical width, its area crossover can reverse that conclusion. The inversion is therefore a way to select and falsify architecture proposals, not a replacement for their detailed simulators.

How a component earns priority. An HPCA mechanism normally reports a local objective such as factory rate, decoder latency, memory density, shuttle count, or gate utilization. QArchGauge does not compare those unlike units directly. It attaches the mechanism to the same record, charges every new event introduced by the mechanism, and asks whether the typed decision changes. At the measured surface-code point, even making decoder or host service free yields less than $1.01\times$ end-to-end speedup, while making factory supply free yields $4,102\times$. A decoder or layout proposal is therefore not rejected as unimportant; it is scheduled after the upstream crossover unless it also reduces rotation supply, shot serialization, or logical work. Conversely, the matched LSQCA experiment shows that a locally denser design can lose application value when its access movement is charged. This gives architects a falsifiable burden of proof: identify the records and operating point that activate the component, replace its matched events, preserve quality and reliability, and demonstrate movement of first target, crossover, successor, or pass count. Reported component speedup alone is insufficient. **Universal FT is one operating point.** The strict physical result assumes surface-code execution and synthesized arbitrary rotations; it does not imply that every useful quantum accelerator should instantiate millions of identical factories. Native or analog rotation schemes can replace part of the T-state term, while neutral-atom and trapped-ion machines expose different transport, reconfiguration, gate-time, and error costs [27], [30]–[32]. Analog Hamiltonian simulators are especially relevant to the eligible Sim. records because their application loop ends in one observable [33], [34]. QArchGauge evaluates such a modality by replacing only the supported rotation, movement, and reliability terms and rerunning the same inversion. The native deadline and quality gate do not move. This distinction matters: specialized hardware may reduce the physical burden for Sim., but it cannot repair the representation and outer-loop failures that block ML, Chem., and Opt. before hardware timing begins.

HPC–QPU co-design. A QPU is an accelerator inside a classical scientific workflow. An integrated runtime should expose the native deadline, finite-shot quality progress, dependency-ready width, factory and decoder backpressure, and host-visible feedback rather than only circuit count. QFw and emerging HPC–quantum ecosystems provide the execution substrate for this contract [21], [35]–[37]. Perlmutter then serves two roles: it produces exact paired evidence before hardware exists, and it remains the competing native path when QPU traces arrive. Replacing a modeled term with a measured QPU trace preserves the comparison instead of rebuilding the benchmark around the new device.

Limits and next evidence. The 4–20-qubit controlled suite prioritizes exact quality closure; the hardware-qualified set is only twelve 4–7-qubit Sim. records. The 36–40-qubit runs are shard-capacity probes, not distributed application demonstrations. Large Chem. lacks a matched VQE loop, Opt. lacks deployment-scale closed-loop QAOA, ML amplitude loading/readout remains unresolved, and no adaptive or mid-

circuit dependency trace is claimed. The FT model also omits factory-internal syndrome traffic, rotation anticommutation depth, measured queue tails, and end-to-end energy. Future work must attach larger quality-closed applications, emitted logical schedules, power traces, and measured QPU output. Energy should compare the complete classical baseline with cryogenics, control, decoding, and factory operation rather than count only device gates [38], [39]. Until then, favorable points are falsifiable architecture targets, not evidence of a realized advantage machine.

VI. RELATED WORK

Application benchmarks and native comparison. SuperMarQ, QASMBench, and MQT Bench characterize applications, circuits, and compilation [2], [40], [41]; QED-C combines quality with time-to-solution, and QUARK compares classical and quantum implementations [3], [12]. A recent Nature perspective likewise asks benchmarks to expose the subsystem they measure [42]. None inverts a measured same-input native deadline into first, crossover, next, and removal-ceiling QPU targets after direct finite-shot loop closure.

FT estimation and critical-path evidence. Beverland et al. compile three scaled applications into size, speed, and controllability requirements [13]; QDK and Qualtran expose QEC, factory, and technology assumptions [26], [43]. TraceQ reconstructs dataflow, d’ArQ groups causal control, and Pinball and decoder studies quantify reaction latency [14]–[16], [24]. QArchGauge adds same-record quality and loop closure plus a measured native deadline before assigning any component target.

QPU component architectures. BOSS, S-SYNC, Cyclone, and LSQCA optimize QCCD movement, memory, or surface-code density [17]–[20]; factory, synthesis, and non-Clifford-state work attacks other terms [29], [44]–[47]. Their gains are mechanism-specific. Our replacement changes only matched events and space, then recomputes the application path to expose both benefit and added movement or reliability cost.

HPC simulation and integration. ScaleQsim, AURORA-Q, and cuQuantum improve distributed state-vector execution [4]–[6]; QFw and laboratory roadmaps study hybrid gateways, scheduling, and CPU/GPU/QPU integration [21], [35]–[37], [48]. QArchGauge instead uses leadership HPC to produce paired evidence and the native competitor, then removes simulator time before QPU projection. It is an architecture diagnosis contract, not a simulator or runtime scheduler.

VII. CONCLUSION

QArchGauge uses leadership HPC as an empirical oracle for future QPU architecture. Its quality-first inversion leaves only 12 same-record, full-loop Sim. targets from 3,552 records. Their strict contracts select distance 13–15 and need 39,351–42,961 $\times$ baseline factory supply before the target moves to useful shot lanes and logical-cycle latency; matched LSQCA movement removes all six otherwise favorable cases. Native-rotation envelopes instead move pressure to readout or 2Q execution. Quantum advantage is therefore a workload- and

phase-specific architecture boundary, not a simulator speedup or one universal hardware factor. A component has demonstrated application value only when it changes the first target, crossover, successor, or pass count under the unchanged quality and reliability contract.

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